

Evaluating Geospatial Foundational Models for Cloud Segmentation: A Comparative Study of Performance, Calibration, and Representations

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Abstract

Clouds are the most obscuring phenomena in geospatial imagery. Their detection as a pre-processing step is paramount for any practical geospatial analysis. In particular, we can leverage geospatial foundational models to handle the task of detection; their generalized feature representations could prove performant and well-calibrated for cloud segmentation. But many factors are available for choosing or designing a geospatial foundational model. Choices of pre-training data (with their modalities), pre-training scheme, and architecture are abundant and unclear in their performance on cloud segmentation. In this work, we elaborate on what foundational model choices make for higher performance and calibration levels. Alongside, we will study ways to estimate foundational model performance and calibration error based on their semantic coherence on cloud sub-phenomena. We find that SAR as a pre-training modality in the data and architecture make no major impact on the performance or calibration of a foundational model for cloud segmentation. The distillation pre-training scheme outperforms reconstructive and contrastive schemes in the out-of-distribution setting. For semantics understanding, seeing past adversarial cloud-shadows was moderately indicative of performance while mutual information between similar cloud sub-phenomena was not indicative of cloud segmentation performance or calibration.

Keywords: Cloud Segmentation, Geospatial Foundational Models, Supervised Learning, Sentinel-2, Landsat 8, SAR

1 Introduction

As destructive phenomena such as forest fires and floods occur, we want to detect their precise locations and predict where future disasters will occur. But clouds deter any geospatial imagery analysis and weaken the validity of results. When working with geospatial imagery such as Landsat 8 or Sentinel-2, images must be free of any issues. The highest processing level accounts for atmospheric effects and geometric alignment, achieving a suitable bottom-of-atmosphere capture of the Earth.

Even so, clouds cannot be easily corrected for. Clouds made of particulate matter gather water vapor and then form. According to NASA, about 66% of the Earth is covered by clouds at any point in time. Since clouds lie far below in the troposphere compared to the satellite orbiting in the thermosphere and exosphere, clouds naturally block radiometric information radiating from the Earth.

To pre-process these clouds, imputation is the ideal procedure for removing these deterrents and

restoring the underlying land information. The detected pixels must be accurately labeled for imputation to be properly carried out, as performing imputation on false positive detections produces misleading land information; these incorrect imputation attempts can obscure results for downstream analysis, even worse so for rare phenomena such as fire burns or flooding. A cloud detection model must output well-calibrated predictions with confidence levels that correspond well to the performance, beyond standard performance, when testing a cloud segmentation model.

To perform segmentation (detection), machine learning solutions dominate the performance in segmentation tasks, as well as many other computer vision tasks. Neural networks are a class of learnable algorithms that contain a highly generic module to soak up patterns (e.g., attention mechanisms or convolutions) and a thresholding component (i.e., an activation function). Two main classifications exist for a neural network: a specialty-trained model and a foundational model. Foundational models have become the choice of model for many tasks, as their generalized features are highly capable in many downstream geospatial tasks. A foundational model is characterized by an architecture trained on a large dataset, with a generic training scheme (typically self-supervised).

When it comes to designing a foundational model, many factors are at play during the pre-training phase before downstream usage. For pre-training a foundational model, a dataset should imbue patterns and biases suitable for any geospatial downstream task. The initial dataset used for the initial pre-training phase is called the pre-training data. As every geospatial imagery dataset attempts to capture the Earth faithfully, every large-scale dataset contains even coverage of the Earth. What truly differentiates these datasets is the modalities available for training a foundational model. In particular, for cloud segmentation, synthetic-aperture radar (SAR) imagery stands as the most competent choice for an additional modality on top of optical imagery, as SAR imagery does not capture clouds due to their large water composition. To take the pre-training dataset and update the weights of the model, a pre-training scheme is required. Modern pre-training schemes are typically self-supervised for their generic optimization goals that are suitable for any particular downstream task. But many

choices are available, such as pixel reconstruction, feature contrastive, or distillation schemes. Each pre-training scheme has completely different means of updating weights, but each scheme demonstrates successful performance across many benchmarks; it is unclear which pre-training schemes are better suited for cloud segmentation. Lastly, the neural network architecture of a foundational model imparts inductive bias that aids in the prediction. The range starts with the attention mechanism of vision transformers that have no inductive bias, to convolutional neural networks that have the most amount of inductive bias.

All of these geospatial foundational model choices potentially make an impact on performance and calibration levels, which stem from their feature representation capabilities. Potentially, we can measure their coherence in understanding cloud sub-phenomena with their features. Cloud shadows are roughly linear displacements of clouds that darken the area, and thin clouds share the same textural aspects of clouds with some level of transparency that admits the underlying land information to pass through.

Many choices are available for a foundational model, but which ones are suitable for performant and well-calibrated cloud segmentation? Also, can we estimate cloud segmentation performance through the coherence of cloud sub-phenomena? We aim to study foundational models through their representations in performance, calibration, and semantic coherence in detecting cloud phenomena. We aim to answer these questions:

- **RQ1:** What foundational model choices are the most performant and well-calibrated?
- **RQ2:** What feature quality measurements correlate with the trained performance and calibration of a foundational model?

Based on these questions, we’ve made the following observations:

- SAR data as a pre-training modality for a geospatial foundational model does not meaningfully enhance the model representations to impact performance or calibration error.
- Distillation pre-training outperforms contrastive and reconstruction pre-training in out-of-distribution performance.

- Foundational model compressive properties of cloud sub-phenomena were not indicative of performance or calibration error.

2 Literature Review

With more specificity, a foundational model is a neural network trained on a large dataset, with a generic training scheme (typically self-supervised). This style of model is designed to extract generalized, meaningful representations to handle any downstream task with a suitable task head. Several factors describe a foundational model: dataset, pretraining scheme, and the architecture. Below, we describe the common choices of parts of current geospatial foundational models. Refer to table 1 for an overview of the geospatial foundational models with their characteristics.

The first factor for a foundational model is **data**. The data sets the underlying bias can have. As the earth is a bounded region and datasets are sampled randomly across the earth, in the limit of data, the capture of the earth by the datasets will naturally converge [1]. What differentiates a dataset is the different modalities that optical imagery cannot capture. The majority of datasets contain purely optical imagery, imagery that contains the visible light spectrum and a large amount of the infrared spectrum; light from the sun strikes the earth, then is reflected off the earth for the satellite to capture the radiometric information. Examples of pre-training datasets are SatlasPretrain [2], Harmonized Landsat-Sentinel (HLS) [3], ISPRS Potsdam [4], and other datasets [5, 6, 7, 8].

Other datasets may include synthetic aperture radar (SAR) imagery; this kind of imagery captures information in the far left side of the electromagnetic spectrum, namely microwaves and radio waves. Imagery here captures structure, rather than surface-level information. Some datasets like SSL4EO-S12 [9], Copernicus-Pretrain, [10], SSL4EO-S12-ML, and other datasets [11, 10, 12] include SAR imagery.

The **architecture** of a model plays a role in the mechanisms used to learn features and representations. Largely, an architecture imbues the level of inductive bias for the model. In computer vision, the locality bias is the strongest bias available for image analysis; it assumes that nearby pixels are more relevant for analysis than farther away pixels. The vision transformer (ViT)

[13] has no inductive bias with the patch-to-patch attention mechanism. On the other end, convolutional neural networks [14] have the strongest level of inductive bias by the convolution operation, which takes a local view of a given pixel location. The rest of the architectures take a middle ground in terms of inductive bias, such as Swin [15], which utilizes the attention mechanism but windows sections of the attention. The majority of geospatial foundational model architectures [16, 17, 18, 8, 11, 19, 20, 21] are vision transformers, especially for later models. Vision transformers are largely chosen for their clean slate of no inductive bias, where researchers prefer to implicitly induce locality bias with large-scale datasets. Some geospatial foundational [2, 22, 23, 24, 6, 7] opt for a non-ViT architecture, either the Swin transformer or the ConvNeXt architecture.

To update the weights of the foundational model, a **pre-training scheme** is required. Besides the traditional supervised approach, such as a classification or segmentation pre-training, modern and more generic techniques exist called self-supervised learning, developed over recent years. Self-supervised learning is a style of unsupervised learning that treats the input itself as a target or label. A traditional example of self-supervised learning is the autoencoder, where the model is trained to recreate the input from some latent representation.

Of geospatial foundational models, there are 4 main flavors of pre-training.

The most common pre-training scheme is the reconstructive scheme, which aims to generatively reconstruct the input through some obscuring transform or perturbation. The commonly used approach is the masked autoencoder scheme [25], which pre-trains a model to recreate partially masked portions of the image. Reconstructive is the widely selected pre-training scheme for geospatial foundational models [22, 21, 6, 20, 11, 10, 8, 18, 16] for their pre-training stability.

Another common pre-training scheme is contrastive learning, a style of featurizing that maximizes agreement between similar feature vectors and potentially maximizes disagreement among dissimilar feature vectors. Contrastive learning can come from a supervised or self-supervised flavor. For the contrasting elements, different facets can be maximized in the contrastive framework; some strategies include maximizing agreement in

seasonality [23], within-patch semantics on a per-patch level (while maximizing disagreement in outside-patch semantics) [16], or their Dynamic World segmentation class distributions [19].

Distillation is another broad category of pre-training that minimizes loss with a teacher network; this teacher network can be another trained neural network, in a traditional distillation training setup [26], or a self-supervised setup where a gradient-delayed teacher network is used through an exponential moving average of the student (trained) network. In the computer vision setting, DINO [27] was the first model to propose distillation as a self-supervised training scheme; many geospatial models [22, 19, 7, 17] implement the DINO style of pre-training. In contrast, others simply use a separately trained encoder to enhance representations [10].

The least-used pre-training scheme in recent geospatial foundational models is the traditional supervised pre-training scheme. Opposed to a single classification task on ImageNet-1K typically done for general computer vision foundational models, supervised pre-trained geospatial foundational models [21, 2, 11] employ a multitude of tasks such as classification, segmentation, object detection, change detection, and regression to ameliorate for the lack of generalized representation learning that self-supervised models perform.

From these foundational model choices, their cloud segmentation performance and calibration are largely untested. We aim to fill the gap in analysis that helps guide decisions for designing models for cloud segmentation.

3 Methodology

In precise terms for our cloud segmentation task, we want an algorithm f_θ that takes in a Landsat 8 or Sentinel-2 image $x \in [0, 2^{12} - 1]^{B \times 224 \times 224}$ (B = number of bands) and outputs a per-pixel prediction $y \in [0, C]^{224 \times 224}$ for C number of classes of at most the background, clouds, thin clouds, and cloud-shadows. The algorithm f should satisfy $f_\theta(x) = y$ for correct pairings of inputs and outputs (x, y) that come from some population distribution.

We compare a selection of foundational models using several criteria that satisfy relevancy and fairness for our cloud segmentation task.

Table 1 Architectures, data sources, modalities, and pre-training schemes by model. Patch size for transformer models is 16×16 unless stated otherwise.

Model	Architecture	Data	SAR	Pre-training schemes			
				Reconstruction	Contrastive	Distillation	Supervised
Prithvi-EO 1.0	ViT [13]	HLS [3]	-	✓	-	-	-
SatlasNet [2]	SwinV2 [15]	SatlasPretrain [2]	-	-	-	-	✓
CMID [22]	SwinV2	ISPRS Potsdam [4]	-	✓	-	✓	-
PIS [23]	SwinV2	SSL4EO-S12 [9]	-	-	✓	-	-
DeCUR [24]	Mit [28]	SUN-RGBD [29]	-	-	✓	-	-
SMLFR [6]	ConvNeXt [14]	GeoSense [6]	-	✓	-	-	-
MTP [21]	ViT (+RVSA)	Million-AID [5], SAMRS	✓	-	-	-	✓
FG-MAE [20]	ViT	SSL4EO-S12 [9]	✓	✓	-	-	-
SoftCon [19]	ViT	SSL4EO-S12-ML [19]	✓	-	✓	✓	-
DINOv3 [7]	ViT-B/14	LVD-1689M [7]	-	-	-	-	-
TerraMind 1.0 [11]	ViT/ConvNeXt	TerraMesh [11]	✓	✓	-	-	✓
CopernicusFM [10]	ViT	Copernicus-Pretrain [10]	✓	✓	-	✓	-
Selective-MAE [8]	ViT	OpticalRS-13M [8]	-	✓	-	-	-
SMARTIES 1.0 [18]	ViT	Mixed [2] [30] [12]	✓	✓	-	-	-
Panopticon [17]	ViT	Mixed [2] [30] [12] [31]	✓	✓	-	✓	-
Galileo [16]	ViT	Mixed	✓	✓	✓	-	-

Table 2 Models grouped by data normalization scheme.

Normalization	Models
Z-score	CopernicusFM, Prithvi-EO 1.0, TerraMind 1.0, DINOv3, Selective-MAE, PIS, CMID, MTP, SMLFR, Panopticon, DeCUR
Z-score w/ 1%/99% percentile clip	SMARTIES 1.0
SatlasPretrain min-max [0,1]	SatlasNet
Min-max w/ 2σ clip (rescaled to [0, 255])	SoftCon, FG-MAE
Min-max w/ 2σ clip	Galileo

- Intended for geospatial tasks.
- Base size models only.
- No vision-language models.
- No hyperspectral models.
- Open-access code and weights.
- Published in 2023 or later.

As the topic is cloud segmentation in Landsat 8 and Sentinel-2 imagery, we only deal with models specified for geospatial imagery. Onto the architecture choices, only base size models ($\approx 100\text{M}$ parameters) are tested because of ubiquity and fairness for comparison. We exclude models that have a language component, as cloud segmentation is performed on a scale too large for language prompts. We also exclude models that deal with hyperspectral bands; drones capture these hyperspectral bands, which are out-of-distribution of the task, as clouds do not meaningfully exist at ground level. All models are listed in table 1.

Foundational models will vary by the bands available from Landsat 8 or Sentinel-2; to remedy the variety of bands, we only use the RGB bands to perform segmentation. We are greatly interested in the underlying spatial semantics of the features, rather than their full final performance. We simply substitute the closest spectral band for the non-RGB bands if required. Bands are normalized by author-recommended normalization settings as listed in table 2.

For the features from the foundational models, some models are capable of outputting multi-scale features, while others are not capable. We bring every model to the same multi-scale capabilities,

augmenting a small ViT-adaptor [32] for single-scale feature models at evenly spaced layers, at layers 3, 6, 7, and 12 for a base ViT. For the multi-scale segmentation task head, we use the standard UperNet [33] head found in many geospatial foundational model comparisons for segmentation. To do the final upscaling, we use DySample [34], which is a lightweight, point-wise upsampler using linear layers with pixel-shuffles.

4 Experiments

4.1 Data

For instrumentation, we chose the two most prolific satellites used for geospatial imagery analysis, Sentinel-2 and Landsat 8.

Sentinel-2 is a 10-meter spatial resolution and 12-bit colorimetric resolution constellation of 2 satellites handled by the European Space Agency (ESA), surveying the same area roughly every 5 days. The instrumentation on Sentinel-2 captures imagery at 10 meters for the RGB bands. Landsat 8 is at a coarser resolution of 30 meters, the same 12-bit color depth, with only a single satellite handled by the United States Geological Survey (USGS), revisiting the same area every 16 days. Landsat 8 carries two different sensors, the operational land imager (OLI) and the thermal infrared-red sensor (TIRS). The OLI instrument captures the RGB bands we are strictly interested in.

L8-Biome [35, 36]: The most widely cited cloud segmentation dataset is the pre-collection Landsat 8 Cloud Cover Assessment dataset, alternatively shorthand as L8-Biome. Collected by the United States Geological Survey (USGS), it is comprised of 96 tiles randomly sampled around the earth at a pixel size of roughly 7000 by 7000. These tiles are selected to evenly sample over the 8 biomes: barren, forest, grass/crops, shrubland, snow/ice, urban, water, and wetlands. The tiles are labeled for clouds, thin clouds, and cloud-shadows; thin clouds are specified as clouds having less than an opacity 50% and cloud-shadows are shadows deriving from a cloud or thin cloud. As this is a pre-collection Landsat 8 dataset, processing quality is subpar compared to the newest collection 1 and 2 processing levels.

Landsat 8 Collection 1 Cloud Cover Assessment (L8-C1-CCA) [37]: Within the cloud

cover assessment family of datasets, the Collection 1 series is a newer dataset collected by the USGS following the same labeling and sampling standards of L8-Biome. The Landsat 8 Collection 1 Cloud Cover Assessment (L8-C1-CCA) comprises 48 tiles randomly selected across the world, labeled for clouds and thin clouds.

CloudSEN12+ [38]: The largest Sentinel-2 cloud segmentation dataset to date is CloudSEN12+, comprising 10,000 randomly sampled tiles sized 509 x 509 pixels and 849 randomly sampled tiles of size 2000 x 2000 pixels, labeled for clouds, thin clouds, and cloud shadows. This dataset is labeled for the same region at multiple timestamps.

Sentinel-2 Cloud Mask Catalogue (S2CMC) [39]: The Sentinel-2 Cloud Mask Catalogue is a selection of 513 images of size 1022x1022 across the earth. Uniquely, this dataset is only labeled for clouds and cloud shadows at 20-meter resolution.

Using these datasets, L8-Biome and CloudSEN12+ are the in-distribution (ID) datasets, meaning the foundational models are trained and tested on these datasets. But L8-C1-CCA and S2CMC are the out-of-distribution (OOD) datasets, never part of the training and validating sessions.

To use all 4 datasets for training, contiguous crops of size 224 x 224 pixels, starting from the top left corner of a tile, are collected and used for training the foundational models. Crops containing NODATA values are excluded from the analysis. With the totality of the crops, we use a 60/15/25 split for training, validating, and testing. We subsample the datasets for computational constraints. To make comparisons between datasets reasonable, we subsample each dataset for 15,000 total crops each as part of the cross-validation process, or 3,750 for testing on the OOD datasets.

4.2 Foundational Models

We select choices for each major axis of a foundational model to compare performance and calibration. On the pre-training dataset modalities, only the SAR modality was frequent enough for a reasonable comparison. SAR imagery contrasts with optical imagery by penetrating through clouds because of their high moisture content. The absence of clouds in SAR imagery and appearance in optical imagery generally detects clouds.

For a geospatial foundational model, having both SAR and optical imagery involved during the pre-training process potentially yields richer representations useful for cloud segmentation, even SAR imagery absent during inference. SAR imagery is not used as input, but marked for inclusion during pre-training. In this setting for SAR, we are studying SAR as a learning under privileged information (LUPI) setting, also known as multi-modal learning with missing modality (MLMM) [40] in the neural network setting. Concretely, we want to investigate if SAR as a pre-training modality can yield downstream performance, SAR being absent during test-time. In the MLMM literature, it is observed that neural networks can benefit from extra-modality during training for deeper feature representation, even if the modality is missing during downstream usage [40]. On the pre-training side, we divide pre-training schemes of the selected foundational models into reconstruction, contrastive, distillation, and supervised as defined in the literature review. For the architecture, vision-transformers are the most represented, while the rest of the architectures are either a convolutional (ConvNeXt), hybrid transformer (Swin), or hierarchical transformer (MiT); we compare whether a model is either a ViT or not, combining ConvNeXt and Swin into a category of architectures that have some level of inductive bias.

4.3 Training

For training, we use a feature extraction approach, freezing the encoder part of the final model and training only the segmentation-specific parts: the ViT-adapter (if present), segmentation head (UPerNet), and upsampler (DySample). For clarity, when we say we train the foundational model, we mean the segmentation task head is being trained. We chose this training approach to study the pure representations that come from the confluence of the underlying pre-training data and the pre-training scheme; further encoder training would shift the representations away from their initial state, losing the influence of pre-training data and scheme.

The full segmentation head is optimized using AdamW [41] with a learning rate of 1×10^{-4} for UPerNet and DySample and 5×10^{-5} for the ViT-Adapter. We use a default weight decay of

($l = .01$) and momentum parameters ($\beta_1 = .99$, $\beta_2 = .999$). For the learning rate, we apply two schedulers: a warm-up scheduler for the first epoch that starts from 10% of the initial learning rate, then smoothly ramps up per step to 100% of the starting learning rate. Then a cosine scheduler is used for the rest of the training, smoothly decreasing the learning rate as the foundational model converges with a minimum learning rate of $\eta_{\min}$ of 1×10^{-6} for the ViT-Adapter and 1×10^{-5} for the rest of the segmentation head with a maximum epochs $T_{\max}$ of 99. A consistent batch size of 32 images is used when the loss is calculated. The loss values 1 are calculated with standard multi-class cross-entropy, with weightings for the classes that are inversely proportional to the square root of the frequency. Training is halted after 5 epochs of minimal change in validation, minimal being a difference of 1×10^{-3} .

$$\mathcal{L}(\{p_i\}_{i=1}^n, \{\hat{p}_i\}_{i=1}^n) = \sum_{i=1}^n \sum_{j=1}^C \frac{1}{\sqrt{f_i}} \cdot p_i^j \log(\hat{p}_i^j)$$

f_i = frequency of class i

4.4 Metrics

For segmentation tasks, intersection-over-union (IoU) is the most relevant metric as it measures the ratio between the correct overlap and the total overlap; the metric will be done per-class, then averaged over the class IoU_j .

$$\text{IoU}_j = \frac{TP_j}{TP_j + FP_j + FN_j} \quad (1)$$

$$TP_j = \text{true positive count of class } j \quad (2)$$

$$FP_j = \text{false positive count of class } j \quad (3)$$

$$FN_j = \text{false negative count of class } j \quad (4)$$

$$\text{IoU} = \frac{1}{C} \sum_{j=1}^C \text{IoU}_j \quad (5)$$

Since segmentation is an upstream task for masking or imputing clouds and their sub-phenomena, a properly calibrated model is paramount. When a reported confidence matches well with the performance, proper downstream decision-making can be done reliably. A practical approach for handling clouds would be imputing

high-confidence predictions while masking low-confidence predictions. Cloud predictions with high certainty can be safely imputed from a well-calibrated model, but predictions from a poorly-calibrated model will lead to erroneous imputations, introducing potentially fatal error into downstream analysis of rare-phenomena detection (e.g., floods and forest fires).

In precise terms, a well-calibrated model's reported confidence values from f_{θ}^p must match the accuracy performance.

$$\mathbb{P}[Y = f_{\theta}(X) \mid f_{\theta}^p(X) = c] = c \quad (6)$$

To assess how well-calibrated a model is, we can take the difference in equation (6) and call this calibration error. To estimate this calibration measurement, we use the typical expected calibration error (ECE) formulation that bins confidence values, then calculates the differences of mean confidence values and accuracy within the bin, and then averages the differences across the bins to estimate how well-calibrated a model is.

$$\text{ECE}_j = \sum_{i=1}^b f_i |\text{Acc}_j^i - \text{Conf}_j^i| \quad (7)$$

$$\text{ECE} = \frac{1}{C} \sum_{j=1}^C \text{ECE}_j \quad (8)$$

$$\text{Acc}_j^i = \text{mean accuracy of class } j \text{ in bin } i \quad (9)$$

$$\text{Conf}_j^i = \text{mean confidence of class } j \text{ in bin } i \quad (10)$$

$$b = \text{number of bins} \quad (11)$$

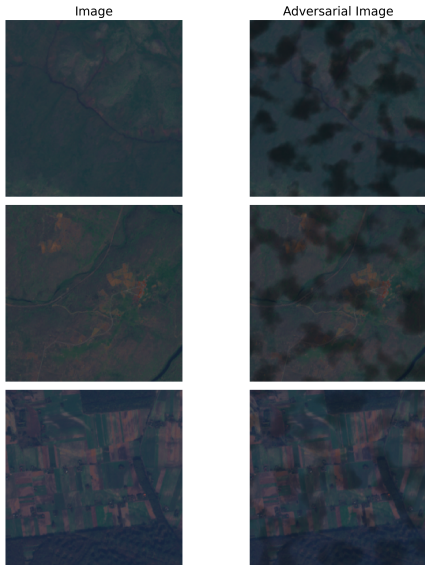
$$f_i = \text{proportion of predictions in the bin } i \quad (12)$$

This metric is calculated class-wise, exactly like for IoU. We chose equal-width ECE as the statistical estimator for calibration error because of its easy interpretability and widespread adoption in the literature.

To understand the semantics of a foundational model, we can elicit the semantic relationship between cloud shadows and the land (background). During training, the segmentation heads only observe cloud shadows accompanied by clouds and/or thin clouds. A foundational model should differentiate shadows stemming from a cloud versus dark patches of the land. That means a model should not classify dark land areas if

Fig. 1 Example images imparted with an adversarial cloud-shadow from the Satellite Cloud Generator [42].

Cloud-shadow perturbation examples



clouds are not present within the image. To form a metric to understand cloud-shadow semantics coherency, we can take the baseline IoU of the model on an image and the difference with the new performance IoU_{adv} , where an adversarial image with darkened areas is planted into the image; the difference will be divided by the baseline IoU to understand the relative error. We call this metric adversarial cloud-shadow error (ACSE); this metric should be as close to 0 as possible. Cloud shadows are generated from the Satellite Cloud Generator [42] package.

$$\text{ACSE} = \frac{\text{IoU} - \text{IoU}_{adv}}{\text{IoU}} \quad (13)$$

To derive a stronger understanding of the class semantics of the foundational models, we can directly measure the overlap in information between the feature vectors of the same region over two time periods with differing classes. Both cloud sub-phenomena, cloud-shadows and thin clouds, transform the land in a way that maintains the relative understandings of the land to be viewable. In a sense, we assume a compression perspective to machine learning, meaning we expect low levels of redundancy from the representations for very similar information. From this

perspective, the information of the same patch between a clear-class timestamp and a cloud-shadow/thin cloud-class timestamp should yield overlapping representations from the feature vectors. To measure the amount of overlapping information, we apply mutual information on the two sets of feature vectors obtained from the foundational model: one set from the unobstructed land, and then another set of features of the same land, but either covered by cloud shadows or thin clouds. In formal terms, mutual information is calculated as the Kullback-Leibler divergence from the outer product of the probability distributions P_X and P_Y to the joint distribution $P_{X,Y}$. The distributions in this case are the distribution of feature vectors from a foundational model of the background class, P_X , and of the cloud-shadow $P_{Y_{cloud-shadow}}$ or the thin cloud class $P_{Y_{thin-cloud}}$.

$$I(X, Y) = D_{KL}(P_{X,Y} \parallel P_X \otimes P_Y) \quad (14)$$

When interpreting mutual information, we want mutual information to be as high as possible in our study, as that indicates a shared representation between the unobstructed land and the land with cloud-shadows/thin clouds. To calculate mutual information, we chose InfoBridge [43] as the estimator of mutual information; this estimator casts estimating mutual information as the movement of marginal distributions $P_X \otimes P_Y$ to the joint distribution $P_{X,Y}$ by solving a stochastic differential equation to transport data-to-data rather than noise to data like typical diffusion models. InfoBridge touts being unbiased, assuming the procedure can be accurately solved with a neural network.

For estimating mutual information, foundation models with bigger feature vector sizes are naturally more capable of packing in more information compared to smaller feature vector dimensions. To achieve fairness among the mutual information comparisons, we scale against the number of dimensions that contributed to the representation of the classes. We can measure this by the effective number of shared dimensions (ENSD [44]). To calculate ENSD, start by defining the feature matrix $X \in \mathbb{R}^{n \times d}$ where n is the number of feature vectors and d is the dimension of the feature vectors to construct the covariance matrix $XX^t \in \mathbb{R}^{n \times n}$. From the covariance matrix, the

participation ratio ρ_X captures the variance of the eigenvectors from the eigenvalues λ_i of the covariance matrix. The participation ratio can be thought of as the number of dimensions used to account for the variance of the distribution of feature vectors.

$$\rho_X = \frac{(\sum_{i=1}^p \lambda_i)^2}{\sum_{i=1}^p \lambda_i^2} = \frac{\text{tr}(XX^t)^2}{\text{tr}((XX^t)^2)} \quad (15)$$

Then ENSD can be defined as the product of the participation ratio of feature matrices X and Y , ρ_X, ρ_Y , and the joint participation ratio τ_{XY} . As ENSD is equivalent to the plain participation ratio of just a single covariance matrix, ENSD can be thought of as a generalization of the participation ratio between two covariance matrices.

$$\text{ENSD} = \rho_X \rho_Y \tau_{XY} \quad (16)$$

$$\tau_{XY} = \frac{\text{tr}((XX^t)(YY^t))}{\text{tr}(XX^t)\text{tr}(YY^t)} \quad (17)$$

The final feature metric is the mutual information density, the amount of mutual information relative to the number of used dimensions for the shared representations of both sets of feature vectors X_{clear} and $Y_{thin-clouds}/Y_{cloud-shadow}$. Mutual information density can only be calculated on the CloudSEN12+, as this dataset is the only one that labels the same area at different timestamps.

$$\text{MI density} = \frac{I(X, Y)}{\text{ENSD}(X, Y)} \quad (18)$$

For all metrics, the mean and standard deviation are reported over 3 seeds. The randomness of performance derives from several factors:

- Module initialization
- Train/validate/test splits
- Batching for AdamW
- Non-determinism of a few CUDA operations

5 Results

5.1 Foundational Model Factors

Of the factors for a foundational model, we start with the pre-training dataset modalities. From table 3, we can observe that for in-distribution IoU performance and calibration error, the addition of the SAR modality was not a consistent or significant benefit. The greatest change was observed in cloud detection IoU performance, but the performance difference displays a mixed bag of results between CloudSEN12+ and L8-Biome. In the out-of-distribution setting, performance and calibration on S2-CMC demonstrated no difference. SAR as a pre-training modality does not demonstrate any improvement in the performance of the representations of the foundational models. Individual model performance can be found in tables A2 and A1.

Between a vision transformer and a non-vision transformer (i.e., ConvNeXt and Swin), no consistent and significant difference could be observed in table 4. Differing levels of inductive bias from neural network architectures did not demonstrate a significant difference in the in-distribution or out-of-distribution IoU performance; the same insignificant difference can be stated for calibration error as well. One aspect about foundational models that potentially explains this non-difference is the large pre-training datasets used for pre-training; a vision transformer would naturally learn the locality bias during pre-training, an inductive bias useful for images.

Observing the performance comparisons of pre-training schemes in table 5, a few noteworthy observations can be made. The distillation pre-training scheme outperformed reconstruction and contrastive in out-of-distribution IoU. The in-distribution setting has mixed results when comparing IoU performance; a consistent and strong pattern could not be found. ECE performance in the in-distribution setting yields minimal differences, as expected, since optimizing cross-entropy naturally promotes accurate and calibrated predictions. When it comes to out-of-distribution ECE, supervised was a better performer against reconstruction and contrastive pre-training, but the mean is too low, and the standard deviation is too high to be a meaningful observation. A sample size of 3 is too small to derive a stable standard

Table 3 Average difference between a model with SAR as pre-training minus a model without SAR as pre-training for IoU and ECE classes (including Overall) within each testing dataset ($n = 3$). Standard deviation is subscripted.

Difference	ID		OOD	
	CloudSEN12+	L8-Biome	S2-CMC	L8-C1-CCA
	IoU ($\uparrow$)	IoU	IoU	IoU
Clear	-0.005 ± 0.002	$+0.001 \pm 0.003$	$+0.004 \pm 0.011$	-0.023 ± 0.004
Cloud	-0.019 ± 0.003	$+0.012 \pm 0.001$	$+0.006 \pm 0.011$	-0.019 ± 0.005
Thin Cloud	$+0.001 \pm 0.008$	$+0.008 \pm 0.003$	—	-0.030 ± 0.006
Cloud-shadow	$+0.002 \pm 0.005$	$+0.003 \pm 0.005$	—	—
Overall	-0.005 ± 0.002	$+0.006 \pm 0.002$	$+0.005 \pm 0.011$	-0.024 ± 0.005
	ECE ($\downarrow$)		ECE	
	ECE	ECE	ECE	ECE
	ECE	ECE	ECE	ECE
Clear	$+0.002 \pm 0.006$	$+0.002 \pm 0.002$	-0.003 ± 0.005	$+0.017 \pm 0.001$
Cloud	-0.002 ± 0.002	$+0.002 \pm 0.002$	-0.003 ± 0.005	$+0.003 \pm 0.001$
Thin Cloud	-0.002 ± 0.006	$+0.002 \pm 0.001$	—	$+0.013 \pm 0.003$
Cloud-shadow	$+0.000 \pm 0.008$	$+0.002 \pm 0.001$	—	—
Overall	-0.001 ± 0.004	$+0.002 \pm 0.000$	-0.003 ± 0.005	$+0.011 \pm 0.002$

Table 4 Average difference of a model with a ViT architecture minus a model without a ViT architecture for IoU and ECE within each testing dataset. Values are mean with subscripted standard deviation ($n = 3$).

Architecture difference	ID		OOD	
	CloudSEN12+	L8-Biome	S2-CMC	L8-C1-CCA
	IoU ($\uparrow$)	IoU	IoU	IoU
ViT vs Non-ViT	-0.002 ± 0.007	$+0.022 \pm 0.002$	-0.000 ± 0.006	-0.029 ± 0.008
	ECE ($\downarrow$)		ECE	
	ECE	ECE	ECE	ECE
	ECE	ECE	ECE	ECE
ViT vs Non-ViT	-0.002 ± 0.007	$+0.001 \pm 0.002$	$+0.003 \pm 0.004$	$+0.014 \pm 0.006$

deviation, leading to the high variances in the out-of-distribution ECE results.

5.2 Indicators of Performance and Calibration

Looking at the ACSE table 6 and Kendall’s τ table 7, coherence of understanding cloud shadows attached to a cloud was a moderate to weak indicator of performance. Even when compared against baseline performance on the image, resistance to adversarial cloud-shadows did not indicate levels of IoU performance or calibration.

Observing table 8 for mutual information density and the table 9 for mutual information density against performance and calibration metrics, mutual information density was largely uncorrelated with the IoU performance and calibration, in both the in-distribution and out-of-distribution setting.

FG-MAE in particular exhibits poor IoU performance and produces high MI density results. The pre-training scheme aims to recreate image features such as histogram of orientated gradients and canny edges, which are granular, low-level information which does not carry high-level semantics. When combining with feature extraction, it results in poor IoU performance. Since these low-level features are high-frequency in nature, a high mutual information is expected, far above the rest of the other foundational models.

6 Conclusion

From the results above for foundational model property choices, SAR pre-training modality, and architecture differences were insignificant in terms of IoU and ECE. The same non-confirming results can be observed for ECE against the different

Table 5 Average pairwise pre-training scheme differences (left scheme minus right scheme) for IoU and ECE within each testing dataset. Values are mean with subscripted standard deviation ($n = 3$).

Pre-training scheme difference	ID		OOD	
	CloudSEN12+	L8-Biome	S2-CMC	L8-C1-CCA
	IoU ($\uparrow$)	IoU	IoU	IoU
Reconstruction vs Contrastive	$-0.012_{\pm 0.003}$	$+0.011_{\pm 0.014}$	$-0.006_{\pm 0.013}$	$+0.016_{\pm 0.016}$
Reconstruction vs Distillation	$-0.035_{\pm 0.004}$	$+0.018_{\pm 0.011}$	$-0.036_{\pm 0.009}$	$-0.047_{\pm 0.012}$
Reconstruction vs Supervised	$+0.004_{\pm 0.009}$	$+0.042_{\pm 0.012}$	$+0.011_{\pm 0.027}$	$-0.019_{\pm 0.030}$
Contrastive vs Distillation	$-0.010_{\pm 0.006}$	$+0.017_{\pm 0.011}$	$-0.010_{\pm 0.008}$	$-0.046_{\pm 0.003}$
Contrastive vs Supervised	$-0.016_{\pm 0.003}$	$-0.002_{\pm 0.013}$	$-0.009_{\pm 0.021}$	$-0.028_{\pm 0.008}$
Distillation vs Supervised	$-0.006_{\pm 0.005}$	$-0.019_{\pm 0.006}$	$+0.001_{\pm 0.025}$	$+0.018_{\pm 0.008}$
	ECE ($\downarrow$)	ECE	ECE	ECE
Reconstruction vs Contrastive	$+0.005_{\pm 0.002}$	$+0.003_{\pm 0.004}$	$+0.003_{\pm 0.014}$	$-0.004_{\pm 0.006}$
Reconstruction vs Distillation	$+0.007_{\pm 0.002}$	$+0.002_{\pm 0.002}$	$+0.006_{\pm 0.013}$	$+0.018_{\pm 0.005}$
Reconstruction vs Supervised	$+0.004_{\pm 0.004}$	$+0.005_{\pm 0.004}$	$+0.014_{\pm 0.028}$	$+0.014_{\pm 0.011}$
Contrastive vs Distillation	$-0.000_{\pm 0.004}$	$-0.001_{\pm 0.002}$	$-0.004_{\pm 0.015}$	$+0.018_{\pm 0.001}$
Contrastive vs Supervised	$+0.001_{\pm 0.005}$	$+0.003_{\pm 0.001}$	$+0.012_{\pm 0.014}$	$+0.010_{\pm 0.005}$
Distillation vs Supervised	$+0.001_{\pm 0.002}$	$+0.004_{\pm 0.002}$	$+0.015_{\pm 0.028}$	$-0.007_{\pm 0.004}$

Table 6 Adversarial cloud shadow error (Overall IoU), mean $\pm$ std ($n=3$ seeds).

Model Name	CloudSEN12Plus		L8-Biome	
	ACSE ($\downarrow$)	Rank	ACSE	Rank
CMID	$0.4086_{\pm 0.1096}$	4	$0.3999_{\pm 0.0846}$	4
Prithvi-EO 1.0	$0.4610_{\pm 0.1492}$	7	$0.5352_{\pm 0.1082}$	13
SatlasNet	$0.1839_{\pm 0.1948}$	2	$0.3255_{\pm 0.0311}$	1
DeCUR	$0.4478_{\pm 0.0536}$	5	$0.4940_{\pm 0.2926}$	8
MTP	$0.6237_{\pm 0.1153}$	17	$0.4779_{\pm 0.2241}$	6
PIS	$0.5084_{\pm 0.1082}$	12	$0.5890_{\pm 0.2427}$	14
SMLFR	$0.4728_{\pm 0.0774}$	10	$0.5000_{\pm 0.1953}$	10
SoftCon	$0.5252_{\pm 0.1041}$	14	$0.4946_{\pm 0.0581}$	9
CopernicusFM	$0.4813_{\pm 0.2889}$	11	$0.3755_{\pm 0.2371}$	2
DINOV3 ConvNeXt	$0.4681_{\pm 0.0228}$	9	$0.8066_{\pm 0.0109}$	17
DINOV3 ViT	$0.5236_{\pm 0.1775}$	13	$0.6692_{\pm 0.0987}$	15
FG-MAE	$0.1700_{\pm 0.2438}$	1	$0.3966_{\pm 0.0672}$	3
Galileo	$0.2505_{\pm 0.1764}$	3	$0.5197_{\pm 0.0691}$	11
Panopticon	$0.5894_{\pm 0.1649}$	16	$0.5349_{\pm 0.1121}$	12
SMARTIES 1.0	$0.5805_{\pm 0.1299}$	15	$0.4914_{\pm 0.1581}$	7
Selective-MAE	$0.4646_{\pm 0.1642}$	8	$0.4349_{\pm 0.1421}$	5
TerraMind 1.0	$0.4482_{\pm 0.1625}$	6	$0.7769_{\pm 0.1347}$	16

pre-training schemes. The only noteworthy result is that distillation pre-training outperforms contrastive and reconstruction pre-training.

In terms of estimating IoU performance and calibration error in both the in-distribution and out-of-distribution settings, adversarial cloud

shadow error was moderately indicative while mutual information density was non-indicative. Potentially, the compression perspective does not explain the performance of foundational models, meaning foundational models may leverage large amounts of redundancy to achieve high

Table 7 Kendall’s τ for cloud-shadow relative error vs IoU, OOD IoU, ECE, and OOD ECE (n=3 seeds).

Domain	Metric	Dataset	Kendall’s τ
ID	IoU	CloudSEN12Plus	0.3824
		L8-Biome	0.2647
	ECE	CloudSEN12Plus	-0.2941
		L8-Biome	-0.1912
OOD	IoU	L8-C1-CCA	0.2500
		S2CMC	0.5441
	ECE	L8-C1-CCA	-0.1765
		S2CMC	-0.3971

performance. In follow-up work, measures of compression could be incorporated as loss calculations in the training of a cloud segmentation model to enhance representations of features, ideally improving upon performance and calibration error.

Though this study is a large comparison of geospatial foundational models for cloud segmentation, a full joint comparison of manually trained models would be the ideal experimentation setup to properly compare foundational model factors; this more thorough study would finely isolate foundational model factors and decision choices for an exact comparison.

We thank the researchers for their open-source code to facilitate this large-scale comparative work on geospatial foundational models for cloud segmentation.

6.1 Reproducibility

All models are trained on a single machine with an AMD 7600X CPU and NVIDIA RTX 5070 Ti. We use 3 fixed seeds: 0, 1, and 2. All reproducibility settings are used for PyTorch and PyTorch Lightning. All sources of randomization are accounted for, besides a select few. ECE calculations are not fully reproducible because the faster frequency counting (`bincount`) implementation does not have a deterministic CUDA implementation. Parts of the segmentation head use a continuous grid sampling function (`grid_sample`), which lacks a deterministic CUDA implementation as well.

Declarations

Author Contributions

C.N. contributed to the conceptualization of the ideas, drafting of the manuscript, and programming. R.V. contributed to supervision and proof-reading.

Conflict of Interest

The authors of this paper do not have conflicts of interest regarding the submission of the work done.

Data Availability

Publicly available:

- Experimentation code is available at <https://github.com/cvnguye3-ncsu/gfm-cloudsegmentation>
- Datasets used are publicly hosted for
 - L8 Biome: <https://landsat.usgs.gov/landsat-8-cloud-cover-assessment-validation-data>
 - L8 C1 Cloud Cover Assessment: <https://www.sciencebase.gov/catalog/item/60536863d34e7e>
 - CloudSEN12Plus: <https://huggingface.co/datasets/tacofoundation/cloudsen12>
 - Sentinel-2 Cloud Mask Catalogue: <https://zenodo.org/records/4172871>

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Appendix A Tabular Results

Tables are provided for the individual IoU and ECE metrics for each model.

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Table 8 Mutual information density mean $\pm$ std (n=3 seeds). Galileo could not be included because of the inclusion of a temporal dimension for a fair comparison.

Model Name	Shadow		Thin Cloud	
	MI Density	Rank	MI Density	Rank
CMID	0.0200 $\pm$ 0.0003	14	0.0173 $\pm$ 0.0011	15
Prithvi-EO 1.0	0.1086 $\pm$ 0.0036	7	0.0974 $\pm$ 0.0055	7
SatlasNet	0.0607 $\pm$ 0.0002	10	0.0705 $\pm$ 0.0022	10
DeCUR	0.0268 $\pm$ 0.0017	13	0.0204 $\pm$ 0.0042	14
MTP	0.1117 $\pm$ 0.0046	6	0.1012 $\pm$ 0.0064	6
PIS	0.2853 $\pm$ 0.0350	2	0.2913 $\pm$ 0.0132	2
SMLFR	0.2256 $\pm$ 0.0085	4	0.1474 $\pm$ 0.0039	5
SoftCon	0.0081 $\pm$ 0.0001	16	0.0078 $\pm$ 0.0006	16
CopernicusFM	0.0583 $\pm$ 0.0091	11	0.0801 $\pm$ 0.0006	9
DINOv3 ConvNeXt	0.0327 $\pm$ 0.0011	12	0.0448 $\pm$ 0.0004	12
DINOv3 ViT	0.0659 $\pm$ 0.0023	9	0.0687 $\pm$ 0.0007	11
FG-MAE	220.1382 $\pm$ 83.3016	1	192.1056 $\pm$ 29.0964	1
Panopticon	0.0185 $\pm$ 0.0012	15	0.0212 $\pm$ 0.0000	13
SMARTIES 1.0	0.2818 $\pm$ 0.0174	3	0.2107 $\pm$ 0.0103	3
Selective-MAE	0.1783 $\pm$ 0.0014	5	0.1664 $\pm$ 0.0031	4
TerraMind 1.0	0.0732 $\pm$ 0.0023	8	0.0855 $\pm$ 0.0033	8

Table 9 Kendall’s τ for MI density vs IoU, OOD IoU, ECE, and OOD ECE (n=3 seeds) for both cloud-shadows/thin clouds.

Domain	Metric	Dataset	Kendall’s τ
ID	IoU	CloudSEN12+	-0.1333 / -0.1000
		L8-Biome	0.1000 / 0.1333
	ECE	CloudSEN12+	0.2167 / 0.1833
		L8-Biome	0.0000 / 0.0000
OOD	IoU	L8-C1-CCA	-0.0667 / -0.1000
		S2CMC	-0.1167 / -0.0833
	ECE	L8-C1-CCA	0.0833 / 0.1167
		S2CMC	-0.0333 / -0.0667

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Table A1 Overall IoU performance ($\mu \pm \sigma$) and within-dataset rank across ID and OOD settings, computed over 3 seeds.

	ID				OOD			
	CloudSEN12+		L8-Biome		S2-CMC		L8-C1-CCA	
Model	IoU ($\uparrow$)	Rank ($\downarrow$)	IoU	Rank	IoU	Rank	IoU	Rank
Prithvi-EO 1.0	0.567 $\pm$ 0.020	16	0.685 $\pm$ 0.013	9	0.716 $\pm$ 0.024	15	0.481 $\pm$ 0.025	16
SatlasNet	0.592 $\pm$ 0.015	12	0.644 $\pm$ 0.009	15	0.737 $\pm$ 0.022	12	0.540 $\pm$ 0.011	11
CMID	0.570 $\pm$ 0.018	14	0.647 $\pm$ 0.041	14	0.711 $\pm$ 0.024	16	0.507 $\pm$ 0.031	12
PIS	0.647 $\pm$ 0.006	5	0.710 $\pm$ 0.014	6	0.782 $\pm$ 0.029	7	0.553 $\pm$ 0.004	9
DeCUR	0.637 $\pm$ 0.013	6	0.657 $\pm$ 0.005	11	0.777 $\pm$ 0.009	9	0.583 $\pm$ 0.016	2
SMLFR	0.632 $\pm$ 0.020	9	0.705 $\pm$ 0.010	7	0.779 $\pm$ 0.005	8	0.584 $\pm$ 0.017	1
MTP	0.622 $\pm$ 0.013	10	0.651 $\pm$ 0.015	13	0.790 $\pm$ 0.003	6	0.574 $\pm$ 0.002	4
FGMAE	0.526 $\pm$ 0.006	17	0.637 $\pm$ 0.007	17	0.703 $\pm$ 0.005	17	0.485 $\pm$ 0.018	15
SoftCon	0.572 $\pm$ 0.003	13	0.644 $\pm$ 0.036	16	0.722 $\pm$ 0.006	14	0.465 $\pm$ 0.011	17
DINOv3 (ViT)	0.668 $\pm$ 0.003	2	0.714 $\pm$ 0.023	5	0.807 $\pm$ 0.013	4	0.555 $\pm$ 0.017	7
DINOv3 (ConvNeXt)	0.633 $\pm$ 0.009	8	0.654 $\pm$ 0.008	12	0.819 $\pm$ 0.022	3	0.574 $\pm$ 0.015	5
TerraMind 1.0	0.686 $\pm$ 0.005	1	0.743 $\pm$ 0.005	1	0.828 $\pm$ 0.029	2	0.548 $\pm$ 0.017	10
CopernicusFM	0.659 $\pm$ 0.001	4	0.698 $\pm$ 0.012	8	0.841 $\pm$ 0.014	1	0.554 $\pm$ 0.019	8
Selective-MAE	0.634 $\pm$ 0.006	7	0.715 $\pm$ 0.011	4	0.757 $\pm$ 0.029	10	0.566 $\pm$ 0.007	6
SMARTIES 1.0	0.568 $\pm$ 0.014	15	0.684 $\pm$ 0.012	10	0.724 $\pm$ 0.002	13	0.502 $\pm$ 0.043	13
Panopticon	0.668 $\pm$ 0.007	3	0.724 $\pm$ 0.007	2	0.800 $\pm$ 0.027	5	0.582 $\pm$ 0.011	3
Galileo	0.616 $\pm$ 0.015	11	0.716 $\pm$ 0.006	3	0.749 $\pm$ 0.015	11	0.493 $\pm$ 0.022	14

Table A2 Overall ECE performance ($\mu \pm \sigma$) and within-dataset rank across ID and OOD settings, computed over 3 seeds.

	ID				OOD			
	CloudSEN12+		L8-Biome		S2-CMC		L8-C1-CCA	
Model	ECE ($\downarrow$)	Rank ($\downarrow$)	ECE	Rank	ECE	Rank	ECE	Rank
Prithvi-EO 1.0	0.038 $\pm$ 0.007	14	0.021 $\pm$ 0.002	14	0.119 $\pm$ 0.002	13	0.103 $\pm$ 0.017	17
SatlasNet	0.031 $\pm$ 0.005	10	0.015 $\pm$ 0.005	6	0.094 $\pm$ 0.025	7	0.068 $\pm$ 0.006	7
CMID	0.043 $\pm$ 0.007	17	0.021 $\pm$ 0.011	13	0.128 $\pm$ 0.024	17	0.084 $\pm$ 0.011	12
PIS	0.028 $\pm$ 0.003	7	0.010 $\pm$ 0.006	1	0.090 $\pm$ 0.019	6	0.075 $\pm$ 0.004	10
DeCUR	0.028 $\pm$ 0.008	9	0.018 $\pm$ 0.001	9	0.113 $\pm$ 0.005	11	0.056 $\pm$ 0.006	1
SMLFR	0.035 $\pm$ 0.016	13	0.018 $\pm$ 0.007	10	0.089 $\pm$ 0.006	4	0.058 $\pm$ 0.007	2
MTP	0.032 $\pm$ 0.009	11	0.020 $\pm$ 0.001	12	0.094 $\pm$ 0.021	8	0.067 $\pm$ 0.005	6
FGMAE	0.043 $\pm$ 0.015	16	0.026 $\pm$ 0.007	17	0.127 $\pm$ 0.025	16	0.087 $\pm$ 0.012	13
SoftCon	0.034 $\pm$ 0.003	12	0.023 $\pm$ 0.005	16	0.122 $\pm$ 0.007	15	0.095 $\pm$ 0.008	15
DINOv3 (ViT)	0.018 $\pm$ 0.002	1	0.012 $\pm$ 0.004	3	0.090 $\pm$ 0.013	5	0.069 $\pm$ 0.006	8
DINOv3 (ConvNeXt)	0.024 $\pm$ 0.005	5	0.017 $\pm$ 0.004	8	0.085 $\pm$ 0.023	3	0.063 $\pm$ 0.002	3
TerraMind 1.0	0.022 $\pm$ 0.007	4	0.011 $\pm$ 0.003	2	0.077 $\pm$ 0.018	2	0.081 $\pm$ 0.004	11
CopernicusFM	0.022 $\pm$ 0.000	3	0.018 $\pm$ 0.002	11	0.057 $\pm$ 0.015	1	0.071 $\pm$ 0.007	9
Selective-MAE	0.028 $\pm$ 0.004	8	0.016 $\pm$ 0.004	7	0.116 $\pm$ 0.025	12	0.064 $\pm$ 0.003	4
SMARTIES 1.0	0.038 $\pm$ 0.003	15	0.023 $\pm$ 0.004	15	0.107 $\pm$ 0.007	10	0.090 $\pm$ 0.023	14
Panopticon	0.020 $\pm$ 0.003	2	0.013 $\pm$ 0.002	5	0.095 $\pm$ 0.017	9	0.065 $\pm$ 0.006	5
Galileo	0.026 $\pm$ 0.011	6	0.013 $\pm$ 0.003	4	0.120 $\pm$ 0.017	14	0.099 $\pm$ 0.009	16

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